

# Advanced Management Accounting

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**Course:** CPA — Certified Public Accountant

**Level:** Advanced Level (CA34S3 — Compulsory Paper)

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**Syllabus version:** Based on the standard KASNEB CPA Advanced Level Advanced Management Accounting syllabus structure — strategic cost management, activity-based costing, performance measurement systems, transfer pricing, and advanced decision-making techniques. Authored from general strategic management-accounting knowledge and standard syllabus topic coverage, not copied from any existing notes provider, textbook, or third-party study guide. Cross-check terminology and emphasis against the current KASNEB syllabus and examiner's reports before the exam.

**Original work notice.** Every explanation, worked example, and question in this document was written from scratch for this paper. No text has been copied or paraphrased from any KASNEB past paper, textbook, or third-party notes/study guide. The scenarios and practice questions are original.

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## Unit overview

**Unit description.** This paper extends Intermediate-level management accounting into strategic territory — using cost and performance information not just to control existing operations, but to actively inform an organisation's competitive strategy. It covers activity-based costing and strategic cost management, modern performance measurement frameworks that go beyond purely financial metrics, transfer pricing between divisions or group entities, and advanced techniques for decision-making under uncertainty.

**Prerequisite.** A candidate attempting this paper is expected to have a secure grasp of Intermediate-level management accounting (cost classification, budgeting, standard costing and variance analysis), since this paper builds directly on those foundations rather than re-teaching them.

**Learning outcomes.** A candidate who passes this paper should be able to:

- Apply activity-based costing to allocate overheads more accurately than traditional absorption costing.
- Evaluate strategic cost management techniques, including target costing and life-cycle costing.

- Design and evaluate a balanced, multi-dimensional performance measurement system.
  - Evaluate transfer pricing methods between divisions or related entities.
  - Apply advanced decision-making techniques under uncertainty, including decision trees and the value of information.
  - Evaluate the strengths and limitations of budgeting as a management control tool, including alternatives such as rolling budgets and beyond-budgeting approaches.
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## 1. Activity-based costing (ABC)

Traditional absorption costing allocates overheads using a single volume-based basis (e.g. machine hours), which can distort product costs when overheads are actually driven by activities unrelated to volume (e.g. the number of production setups, or purchase orders processed). **Activity-based costing** instead identifies distinct **cost pools** for major activities (e.g. machine setups, quality inspections, order processing), each with its own **cost driver** (a measure that genuinely causes the cost to be incurred, e.g. number of setups), and allocates overhead to products based on how much of each cost driver a product actually consumes. ABC generally produces more accurate product costs where products consume overhead-generating activities in significantly different proportions to their production volumes — for instance, a low-volume, highly customised product that requires frequent setups may be significantly under-costed by traditional volume-based absorption, since its setup-driven overhead gets diluted across the whole production volume rather than attributed specifically to it.

## 2. Target costing and life-cycle costing

**Target costing** works backward from the market: starting with a competitive **target selling price** the market will bear, subtracting the company's **required profit margin**, to arrive at a **target cost** the product must be designed and produced within — reversing the traditional "cost-plus" logic of first determining cost, then adding a margin to set price. Where a product's currently estimated cost exceeds its target cost, the company must find ways to close the gap (value engineering, redesign, supplier negotiation) before launch, rather than accepting a lower margin. **Life-cycle costing** looks beyond a single period's production cost to consider the *total* cost of a product across its entire life cycle — including pre-production costs (research and development, design) and post-production costs (customer support, warranty, disposal) — since a product with a low unit production cost but very high post-sale support costs may actually be less profitable overall than its production cost alone would suggest.

## 3. The balanced scorecard and non-financial performance measures

Purely financial performance measures (profit, ROI) tend to be **lagging indicators** — they report what has already happened, often too late to correct an emerging problem. The **balanced scorecard** framework addresses this by measuring performance across four linked perspectives: the **financial perspective** (how do we look to shareholders?), the **customer perspective** (how do customers see us?), the **internal business process perspective** (what must we excel at internally?), and the **learning and growth perspective** (can we continue to improve and create value?). The underlying logic is that improvements in the non-financial perspectives (e.g. employee capability, process efficiency, customer satisfaction) are genuine **leading indicators** that drive future financial performance — giving management earlier warning and more actionable levers than financial results alone would provide.

## 4. Divisional performance measurement — ROI and residual income

**Return on Investment (ROI)** = Divisional profit ÷ Divisional net assets (investment), expressed as a percentage — intuitive and comparable across divisions of different sizes, but can create a perverse incentive: a divisional manager whose division already earns a high ROI may reject a new project with a perfectly acceptable return (above the company's cost of capital) simply because it would slightly dilute their division's already-high average ROI.

**Residual income (RI)** = Divisional profit – (Imputed interest charge × Divisional net assets), where the imputed interest charge reflects the company's required rate of return on capital employed. RI avoids ROI's perverse incentive, since any project earning above the

company's required rate of return will increase RI and should therefore be accepted by a manager evaluated on RI, aligning divisional incentives more closely with overall company wealth maximisation.

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## 5. Transfer pricing

**Transfer pricing** is the price charged when one division or group entity sells goods/services to another related division or entity. Common methods include: **market-based transfer pricing** (using the external market price, where one genuinely exists, generally considered the most economically sound basis since it reflects a true opportunity cost), **cost-based transfer pricing** (based on the selling division's cost, plus a margin — simpler but can distort incentives if it doesn't reflect genuine opportunity cost), and **negotiated transfer pricing** (divisions negotiate a mutually acceptable price, useful where no reliable external market price exists, though it depends heavily on the divisions' relative negotiating positions). Where divisions operate across international borders, transfer pricing also carries significant **tax implications** — since shifting profit toward a lower-tax jurisdiction via transfer pricing is closely scrutinised by tax authorities, requiring transfer prices to be set on an "arm's length" basis reflecting what unrelated parties would genuinely agree to.

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## 6. Decision-making under uncertainty — decision trees

A **decision tree** maps out a sequence of decisions and the uncertain outcomes (with associated probabilities and payoffs) that follow each decision, allowing a decision-maker to compute the **expected value** of each decision path (working backward from the final outcomes) and select the path with the highest expected value. Decision trees are particularly useful where a decision involves multiple sequential stages (e.g. an initial decision to test-market a product, followed by a subsequent launch decision that depends on the test result), since they force explicit, structured consideration of exactly how later decisions depend on the outcomes of earlier ones.

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## 7. The value of information

Before commissioning additional market research or further analysis, it's worth explicitly evaluating whether that information is actually worth its cost. The **expected value of perfect information (EVPI)** is the difference between the expected value of a decision made with perfect foresight of which outcome will actually occur, and the expected value of the best decision made without that additional information — representing the maximum a

rational decision-maker should be willing to pay for genuinely perfect information. Since perfect information is rarely actually available, the **expected value of sample information (EVSI)** — evaluating a specific, realistically available research or survey option — is generally the more practically relevant calculation, and should always be compared against the actual cost of obtaining that information before commissioning it.

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## 8. Alternatives to traditional budgeting

Traditional annual budgeting has drawn genuine criticism: it can become outdated quickly in a fast-changing environment, can encourage short-term, target-driven behaviour that doesn't serve the organisation's genuine long-term interests, and consumes significant management time on a process that may add limited real strategic value. **Rolling budgets** address the "quickly outdated" problem by continuously extending the budget forward (e.g. always maintaining a twelve-month forward view, adding a new month as each month passes) rather than fixing the whole year in advance. The **beyond budgeting** movement goes further, arguing organisations should replace traditional fixed annual budgets entirely with more adaptive, rolling forecasts and relative performance targets (benchmarked against peers or prior periods, rather than a single fixed number set far in advance) — a genuinely significant change in management philosophy that not every organisation is willing or able to adopt.

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## 9. Full worked question — ABC versus traditional overhead absorption

**Scenario.** A company produces two products, X (high volume, simple) and Y (low volume, complex, requiring frequent machine setups). Total overheads of KSh 1,000,000 relate to machine setups, driven by the number of setups. Product X requires 20 setups; Product Y requires 80 setups (100 setups in total). Under traditional absorption costing, overhead is absorbed based on machine hours: X uses 8,000 machine hours, Y uses 2,000 machine hours (10,000 total).

**Required:** (a) Compute the overhead allocated to each product under traditional (machine-hour based) absorption costing. (b) Compute the overhead allocated to each product under activity-based costing (based on number of setups). (c) Comment on the difference.

**Solution:**

(a) Traditional: overhead rate =  $1,000,000 \div 10,000$  machine hours = KSh 100 per machine hour.

Product X =  $8,000 \times 100$  = **KSh 800,000**. Product Y =  $2,000 \times 100$  = **KSh 200,000**.

(b) ABC: overhead rate =  $1,000,000 \div 100$  setups = KSh 10,000 per setup.

Product X =  $20 \times 10,000 = \text{KSh } 200,000$ . Product Y =  $80 \times 10,000 = \text{KSh } 800,000$ .

(c) Traditional absorption costing significantly **under-costs** Product Y (allocating only KSh 200,000, versus KSh 800,000 under ABC) and significantly **over-costs** Product X, because it allocates overhead based on machine hours, while the overhead is actually driven by the number of setups. Since Product Y is low-volume but setup-intensive, ABC reveals that it consumes far more of the setup-related overhead than its machine-hour usage alone would suggest — a distortion that could lead management to under-price Product Y or over-estimate Product X's true profitability under the traditional method.

## 10. Practice examination — KASNEB-style paper (original, not a past paper)

**Note on format.** This section is laid out the way a KASNEB CPA examination paper is laid out — a timed paper with five compulsory 20-mark questions — purely so you can practise under realistic exam conditions. Every question below is an original scenario written for this document. **It is not a reproduction of any actual KASNEB past examination paper.**

### CPA ADVANCED LEVEL — ADVANCED MANAGEMENT ACCOUNTING (PRACTICE PAPER)

**TIME ALLOWED: 3 HOURS**

*Instructions to candidates: Answer ALL FIVE questions. Each question carries 20 marks. Show all workings clearly.*

#### QUESTION ONE

- (a) Explain why traditional volume-based overhead absorption can distort product costs. (8 marks)
- (b) Define "cost pool" and "cost driver" in the context of activity-based costing. (6 marks)
- (c) Outline ONE circumstance in which ABC is likely to produce materially different product costs from traditional absorption costing. (6 marks)

*(Total: 20 marks)*

#### QUESTION TWO

- (a) Explain target costing, distinguishing it from traditional cost-plus pricing. (10 marks)

(b) Explain life-cycle costing and why a product's production cost alone can understate its true total cost. (10 marks)

(Total: 20 marks)

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### QUESTION THREE

(a) Outline the FOUR perspectives of the balanced scorecard. (12 marks)

(b) Explain why non-financial measures are described as "leading indicators" relative to financial measures. (8 marks)

(Total: 20 marks)

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### QUESTION FOUR

A division has profit of KSh 4,000,000 and net assets of KSh 20,000,000. The company's required rate of return is 12%.

(a) Compute the division's ROI. (4 marks)

(b) Compute the division's residual income. (6 marks)

(c) Explain why a divisional manager evaluated on ROI might reject a new project earning 14%, even though it exceeds the company's required rate of return, and explain why residual income avoids this problem. (10 marks)

(Total: 20 marks)

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### QUESTION FIVE

(a) Distinguish between market-based, cost-based and negotiated transfer pricing, noting one advantage or disadvantage of each. (15 marks)

(b) Explain why cross-border transfer pricing attracts particular scrutiny from tax authorities. (5 marks)

(Total: 20 marks)

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## 11. Marking guide — model answers to the practice examination

### QUESTION ONE

(a) (8 marks) — Volume-based absorption spreads overhead in proportion to volume (e.g. machine hours), which can misrepresent true cost where overheads are actually driven by non-



volume-related activities (e.g. number of setups), over-costing high-volume simple products and under-costing low-volume complex ones.

(b) (6 marks) — Cost pool: a grouping of overhead costs related to a specific activity. Cost driver: a measure that genuinely causes that cost to be incurred (e.g. number of setups).

(c) (6 marks) — Where a low-volume, complex product consumes overhead-generating activities (e.g. setups) disproportionately to its production volume, ABC and traditional absorption will produce materially different, and traditional absorption will materially understate its true cost.

## QUESTION TWO

(a) (10 marks) — Target costing starts from a market-determined target price, subtracts required margin to derive a target cost the product must be designed/produced within; cost-plus pricing instead starts from cost and adds a margin to set price — target costing reverses this logic, driven by market price first.

(b) (10 marks) — Life-cycle costing considers total cost across a product's entire life (R&D/design, production, post-sale support/warranty/disposal), not just production cost; a product with low production cost but high post-sale support costs can be less profitable overall than production cost alone suggests.

## QUESTION THREE

(a) (12 marks — 3 marks each) — Financial, customer, internal business process, learning and growth.

(b) (8 marks) — Improvements in non-financial measures (e.g. employee capability, process efficiency, customer satisfaction) tend to drive future financial performance, giving management earlier warning and more actionable levers than financial results, which only report what has already happened.

## QUESTION FOUR

(a) (4 marks) —  $\text{ROI} = 4,000,000 \div 20,000,000 = \mathbf{20\%}$ .

(b) (6 marks) —  $\text{Imputed interest} = 20,000,000 \times 12\% = 2,400,000$ .  $\text{RI} = 4,000,000 - 2,400,000 = \mathbf{\text{KSh } 1,600,000}$ .

(c) (10 marks) — A project at 14% would lower the division's already-high 20% average ROI even though 14% exceeds the company's 12% required return, creating a perverse incentive to reject it under ROI evaluation; under RI, the project would add  $(14\% - 12\%) \times \text{investment}$  as positive residual income, correctly incentivising acceptance of any project earning above the required rate.

## QUESTION FIVE



(a) (15 marks — 5 marks each) — Market-based: uses external market price, generally most economically sound where one exists, but not always available. Cost-based: uses selling division's cost plus margin, simpler but can distort incentives if it doesn't reflect true opportunity cost. Negotiated: divisions agree a mutually acceptable price, useful absent a reliable market price, but depends on relative negotiating power.

(b) (5 marks) — Cross-border transfer pricing can be used to shift profit toward lower-tax jurisdictions, so tax authorities scrutinise it closely, requiring prices to be set on an arm's-length basis reflecting what unrelated parties would genuinely agree to.

## 12. Exam technique and revision checklist

- ABC questions: always compute the ABC-based cost driver rate and traditional volume-based rate explicitly, then explicitly compare and interpret the difference, as the worked example above demonstrates.
- Target costing vs cost-plus: anchor your answer to which comes first — market price (target costing) or cost (cost-plus) — as the defining distinction.
- Balanced scorecard: know all four perspectives by name and be ready to give an example measure for each.
- ROI vs residual income: always explain the specific perverse incentive ROI creates, and why RI's imputed-interest-charge structure avoids it.
- Transfer pricing questions: pair each method with a genuine advantage/disadvantage rather than a bare definition, and always mention the tax/arm's-length angle for cross-border scenarios.

### Quick revision — one line per topic:

1. ABC: cost pools + cost drivers allocate overhead more accurately than a single volume-based basis.
2. Target costing: target price – required margin = target cost (reverse of cost-plus).
3. Life-cycle costing: total cost across R&D, production and post-sale support/disposal.
4. Balanced scorecard: financial, customer, internal process, learning and growth.
5. ROI = Divisional profit ÷ Net assets; can create a perverse incentive against good projects.
6. Residual income = Profit – (Imputed interest × Net assets); avoids ROI's perverse incentive.
7. Transfer pricing: market-based (most sound where available), cost-based, negotiated.
8. Decision trees: map sequential decisions/outcomes, select highest expected value path.
9. EVPI: max worth paying for perfect information; EVSI: more realistic, for actual available research.

10. Rolling budgets/beyond budgeting: adaptive alternatives to fixed traditional annual budgeting.

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*End of study notes. This document is intended as original exam-preparation material and should be used alongside the current official KASNEB syllabus and examiner's reports for the paper.*